Emile Takahiro Okada

Curriculum Vitae

Nanyang Technological University, Singapore
emile.okada@ntu.edu.sg emileokada.com

Personal Particulars

- Nationality: Norwegian.

Employment History

- Assistant Professor (Tenure-Track), *Nanyang Technological University.* 2026–present
- Peng Tsu Ann Assistant Professor, *National University of Singapore.* 2022–2026

Education

- DPhil in Mathematics, *University of Oxford* 2018–2022
  Thesis: *The Wavefront Set of Representations of p-adic Groups*
  Supervisors: Dan Ciubotaru and Kevin McGerty
- Master of Mathematics, *University of Cambridge* 2017–2018
  Distinction
  Thesis: *Modular Representations of Finite Groups of Lie Type in the Defining Characteristic*
- BA in Mathematics, *University of Cambridge* 2014–2017
  First Class in all years

Research

Joint papers use alphabetical author order.

Preprints

10. *Microlocal Arthur packets as a quasi-canonical basis of stable distributions* (with D. Ciubotaru), 2026. arXiv:2610.16931.
9. *A Microlocal Description of Aubert-Zelevinsky Duality on Unipotent L-Parameters* (with J. Antor), 2026. arXiv:2605.06180.
8. *Ramification of weak Arthur packets for p-adic groups* (with M. Gurevich), 2024. arXiv:2404.03485.

Publications

7. *Types and stable wave front set of the theta representation* (with E. Karasiewicz and R. Wang). *Advances in Mathematics* **503**, Part C (2026), article 111240.
6. *Local character expansions and asymptotic cones over finite fields* (with D. Ciubotaru). *Proceedings of the London Mathematical Society* **130** (2025), no. 5, e70046.
5. *Wavefront Sets of Unipotent Representations of Reductive p-adic Groups II* (with D. Ciubotaru and L. Mason-Brown). *Journal für die reine und angewandte Mathematik (Crelle's Journal)* **823** (2025), 191–253.

4. *Some Unipotent Arthur Packets for Reductive p-adic Groups* (with D. Ciubotaru and L. Mason-Brown). *International Mathematics Research Notices* (2024), no. 9, 7502–7525.

3. *The Wavefront Sets of Unipotent Supercuspidal Representations* (with D. Ciubotaru and L. Mason-Brown). *Algebra & Number Theory* **18** (2024), no. 10, 1863–1889.

2. *Wavefront Sets of Unipotent Representations of Reductive p-adic Groups I* (with D. Ciubotaru and L. Mason-Brown). Accepted to *American Journal of Mathematics*. arXiv:2112.14354.

Thesis

1. *The Wavefront Set of Representations of p-adic Groups*. DPhil thesis, University of Oxford, 2022.

Research Funding

- Nanyang Assistant Professorship, *Nanyang Technological University.* 2026
- Conference funding for *Wavefront Sets in Number Theory and Representation Theory*, Institute for Mathematical Sciences, National University of Singapore. 2027

Awards and Honours

- Newton International Fellowship, Royal Society (£280,000; awarded; declined). 2025
- Aker Scholarship (doctoral studies, University of Oxford). 2018–2022
- Cambridge University Trust Scholarship (undergraduate and master's studies). 2014–2018
- Examination Prize, Emmanuel College, Cambridge (four awards, one each year). 2015–2018

Invited Talks

30 invited conference and seminar talks, including at *Massachusetts Institute of Technology, University of Cambridge, University of Oxford, Tsinghua University, Peking University, University of Bonn, Weizmann Institute of Science, Mathematisches Forschungsinstitut Oberwolfach.*

Invited Conference Talks

- BIRS, Hangzhou, China, *Wavefront Sets of Representations and Endoscopic Classification.* Oct 2026
- Institute of Mathematics of University of Sevilla, Spain, *The Unitary Dual.* Jun 2026
- Mathematisches Forschungsinstitut Oberwolfach, Germany, *Representations of p-adic groups.* Dec 2024
- Vietnam Institute of Advanced Study in Mathematics, Vietnam, *Pan Asian Number Theory Conference.* Jul 2024
- Australian National University, Australia, *Pacific Rim Conference in Mathematics.* Jun 2024
- Institute for Advanced Study in Mathematics, Hangzhou, China, *Arthur Packets.* Nov 2023

- Nisyros, Greece, *Anaparastaseis: Orbits, Hecke algebras, and representations.* Jul 2023
- National University of Singapore, Singapore, *Representation Theory, Combinatorics and Geometry.* Dec 2022
- Erwin Schrödinger International Institute for Mathematics and Physics, Austria, *Minimal Representations and Theta Correspondence.* Apr 2022
- Durham University, Durham, UK, *Representations of p-adic groups and related topics.* Sep 2021

Other Conference Participation

- Brin Mathematics Research Center, United States of America, *Representation Theory and Lie Groups.* Oct 2024
- Centre International de Rencontres Mathématiques, Marseille, France, *Buildings and Affine Grassmannians.* Aug 2019

Teaching

- *Lecturer, National University of Singapore*
  - Introduction to Analytic Number Theory (MA4263). 2026
  - Lie Theory (MA5211). 2022, 2025
  - Algebra II (MA3201). 2023
- *Tutor, Somerville College, University of Oxford*
  - Prelims Analysis II and III; Part A Integration, Topology, and Groups. 2021
- *Teaching Assistant, Mathematical Institute, University of Oxford*
  - Courses in representation theory, commutative algebra, Galois theory, homological algebra, and category theory. 2018–2022

Mentorship

- *PhD students*
  - Yikun Fan (NTU) 2026–present
- *Master's students*
  - Timothy Peck Hsien Wei (NUS) 2023–2024
- *Undergraduate research students*
  - Jerome Quek (NTU, undergraduate research programme) 2026–2027
  - Phyrus Wong Hong Xiang (NTU, final-year project) 2026–2027
  - Malcolm Tan (NUS) 2026
  - Galileo Grey (NUS) 2024–2025
- *Research mentoring*
  - Joven Teo Jia Xuan (PhD student, NUS) 2025–present

Service

- *Conference co-organiser*: Wavefront Sets in Number Theory and Representation Theory, Institute for Mathematical Sciences, National University of Singapore. Sep 2027
- *Referee and quick opinions for*: *Forum of Mathematics, Pi*; *Proceedings of the London Mathematical Society*; *Journal für die reine und angewandte Mathematik (Crelle's Journal)*; *Selecta Mathematica*; *Advances in Mathematics*; *Algebra and Number Theory*; *International Mathematics Research Notices*.
- *Seminar Organiser*: p-adic Arthur packets (2024), The local Langlands correspondence for unipotent representations (2022), Algebraic groups (2019).

Other Employment

- Stipendiary Lecturer, *Somerville College, University of Oxford*. Jan–Sep 2021
- Foreign Exchange Intern, *DNB Bank ASA*. Jul–Aug 2018
- Digital Innovation and Business Development Intern, *DNB Bank ASA*. First place, company-wide hackathon. Jun–Sep 2017
- Summer Undergraduate Research Project (SUROP), Cambridge Image Analysis Group, *University of Cambridge*. Jun–Aug 2016

References

- Dan Ciubotaru, *University of Oxford*, dan.ciubotaru@maths.ox.ac.uk.
- Wee Teck Gan, *National University of Singapore*, matgwt@nus.edu.sg.
- Ju-Lee Kim, *Massachusetts Institute of Technology*, juleekim@mit.edu.
- Maxim Gurevich, *Technion - Israel Institute of Technology*, maxg@technion.ac.il.
- Lucas Mason-Brown, *The University of Texas at Austin*, lucas.masonbrown@gmail.com.